

Contact

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www.linkedin.com/in/khisal
(LinkedIn)

Top Skills

Gazebo

Robot Operating System (ROS)

Research and Development (R&D)

Languages

English (Native or Bilingual)

Urdu (Native or Bilingual)

German (Limited Working)

Punjabi (Native or Bilingual)

Certifications

DeepLearning.AI TensorFlow
Developer Specialization

Certificate for attending workshop on
Arduino

Certificate of Participation for
attending workshop on robotics, SSL
vision software & basic strategies for
Robosprint 2014

Crash Course on Python by Google

Introduction to TensorFlow for
Artificial Intelligence, Machine
Learning, and Deep Learning by
deeplearning.ai

Publications

Pick and Place Multi-Axis Arm for
Object Sorting Based on Color
Sensor",

Muhammad Khisal Khalid

Explainable AI | Computer Vision | Sensor Fusion | Robotics (ROS2)
| Project Management | Development Engineer @ marinom |
Maritime Autonomy

Bremen, Bremen, Germany

Summary

I build AI systems that operate in the real world, not just in simulation. My work sits at the intersection of Computer Vision, Sensor Fusion, Robotics (ROS2), and Explainable AI, with a focus on safety-critical autonomous applications.

At marinom, I lead the end-to-end technical development and project management of a BMFTR-funded collaborative research project with academic and industrial partners. I am building a real-time situational awareness and navigation assistance system from the ground up, covering multi-sensor fusion, regulation-compliant decision support, and Explainable AI for safety-critical decision logging. Alongside the technical work, I handle consortium coordination, milestone reporting, and BMFTR deliverables across multiple work packages.

Earlier in my career, I worked in aerial robotics on multi-target detection with RGB/thermal imaging on Nvidia Jetson and ROS, and contributed to academic research across Brain-Computer Interfaces, agricultural imaging, and synthetic dataset generation.

My background spans about ten years of combined academic and industry experience in Germany, with a BS in Electrical Engineering and an MSc in Control, Microsystems and Microelectronics.

Experience

marinom GmbH

Development Engineer

October 2023 - Present (2 years 9 months)

Bremen, Germany

Explainable AI(xAI), Computer Vision, Robotics, Machine Learning, Project Management, ROS, Sensor Fusion, AI Ethics

Wargdrones

Machine Learning and Computer Vision Engineer

March 2021 - August 2023 (2 years 6 months)

Oldenburg, Lower Saxony, Germany

Developing Machine Learning models for multiple drone-based detections

Responsible for the people detection pipeline

Working with multi-sensor image data (RGB and Thermal) for computer vision

Nvidia Jetson-based Development

Utilized drone imagery to generate point clouds to programmatically establish no-fly zones

Object Detection and Tracking

2D and 3D key point estimation, surface reconstruction and depth estimation

SFB EASE

Student Research Assistant in Computer Vision and Machine Learning

April 2020 - February 2021 (11 months)

Bremen, Germany

-Applying Machine Learning for Computer Vision experiments.

-Statistical Analysis of experimental data in Python.

-Developing rendering components to generate imagery of textured 3D models.

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)

2 years 1 month

Masters Project

November 2018 - August 2020 (1 year 10 months)

Bremen Area, Germany

-Generation of Synthetic RGBD datasets using the technique of Domain Randomization.

-Training of Synthetic Datasets Using TensorFlow for the purpose of object detection

-Study the behaviour and accuracy of datasets by training through Tensorflow object detection API.

Student Research Assistant

August 2018 - March 2019 (8 months)

Bremen, Germany

University of Bremen

Student Research Assistant

December 2019 - February 2020 (3 months)

Bremen Area, Germany

- Maintain and improve eye tracking visualization software
- Work on the visual interface of the software
- Prepare scientific material to be used in scientific publications

BIBA - Bremer Institut für Produktion und Logistik GmbH

Student Research Assistant

January 2018 - October 2019 (1 year 10 months)

Bremen Area, Germany

- Processing application-oriented research questions of image processing in the production and logistics area
- Active cooperation with industrial partners in project consortia
- Development of new methods and concepts for object recognition and quality control

Work Domain: Computer Vision, Python, Artificial Intelligent and Machine Learning

Lahore University of Management Sciences

Research Assistant

January 2016 - March 2017 (1 year 3 months)

Lahore

- Precision Agriculture/Plant management system

Created a MATLAB filter to acquire valuable information from Plants imagery like Greenness index, Biomass, NDVI using image processing techniques.

Platform Used: MATLAB Image Processing Toolbox

- Forest Management

Successfully developed MATLAB algorithms which detect and count number of trees in both satellite and drone images.

Platform Used: MATLAB Image Processing Toolbox

- Biomechanics Research Lab

I major tasked were to analyze the bowling action of the cricket players using different research models. To develop localization algorithms on robots such as quadcopters.

Platform Used: Vicon Motion Capture System, Nexus, MATLAB

- Brain Computing Interface

I gave technical assistance to detect and manipulate brainwaves for the control of RC car and 5 axis robotic arm.

Platform Used: OpenVibe, Arduino

- Teaching Assistant

For the course, Introduction to Artificial Intelligence. My responsibilities include invigilation, helping students out with practice questions in class and managing projects.

Platform Used: Python and MATLAB for student projects

BHANERO TEXTILE MILLS LTD

Internship

July 2015 - December 2015 (6 months)

18km Faisalabad road Sheikhpura

Pak Elektron Limited

Summer Intern

June 2014 - July 2014 (2 months)

Lahore Pakistan

Education

University of Bremen

Master of Science - MS, Control, Microsystems,
Microelectronics · (2017 - 2021)

University of Central Punjab

Bachelor of Science (BS), Electrical and Electronics
Engineering · (2011 - 2015)

Punjab Group Of Collages

Intermediate in science, FSc Pre Engineering · (2009 - 2011)